

Computer Science and Engineering Department, SVNIT, Surat
M.Tech. I – Semester I
Design and Analysis of Algorithm (CSDS103)
Lab Assignment 3
Dynamic Programming

General Instructions

1. Develop the algorithm and identify the subproblem/state before implementing the program.
2. Provide the recurrence or state transition used in the solution.
3. Analyse the time and auxiliary space complexity of the proposed algorithm.
4. Test the solution using suitable inputs, including boundary cases, and justify the correctness of the approach.

Problem 1 – Coin Change

You are given a set of coin denominations and a target amount. Each denomination can be used any number of times. Determine the **minimum number of coins** required to obtain the target amount. If the target amount cannot be formed using the given denominations, report that no solution exists.

For example, for the denominations $\{1, 3, 4\}$ and target amount 6, the minimum number of coins is 2, using $3+3$.

Your program should also display one corresponding combination of coins.

1. Formulate the problem using Dynamic Programming by identifying an appropriate state and recurrence.
2. Implement the solution using a bottom-up Dynamic Programming approach.
3. Compare the DP solution with a straightforward recursive solution in terms of time and auxiliary space complexity.
4. Experiment with different numbers of denominations and target amounts and observe how the execution time changes.

Analysis:

Consider a case where a greedy strategy chooses the largest possible denomination at every step. Determine whether the greedy strategy always produces the minimum number of coins. If not, provide a counterexample and explain why Dynamic Programming succeeds where the greedy approach fails.

Problem 2 – Employee Training Planning

An organisation needs to train an employee in a sequence of training modules. Each module has a training cost and provides a certain benefit to the employee.

The employee has a maximum training budget of B . From the available modules, each module can either be selected once or skipped. The objective is to select a set of modules whose total cost does not exceed B while maximising the total benefit.

For example, consider the following modules:

Module	Cost	Benefit
M_1	4	7
M_2	6	10
M_3	5	8
M_4	3	5

Given n training modules and a maximum budget B , determine the maximum achievable benefit and identify the selected modules.

1. Formulate the problem using Dynamic Programming by identifying the state and state transition.
2. Implement the solution and reconstruct the selected set of training modules.
3. Analyse the time and auxiliary space complexity of your solution.
4. Study the effect of increasing the number of modules and the available budget on the execution time.

Analysis:

Consider the following modification: a training module may now be selected more than once. Determine how this changes the problem formulation and the Dynamic Programming solution. Explain the difference between the original and modified problem and analyse the effect on the recurrence and complexity.